

Lecture 13: KKT Second-Order Conditions for Constrained Optimization Problems

MA3236 Non-Linear Programming

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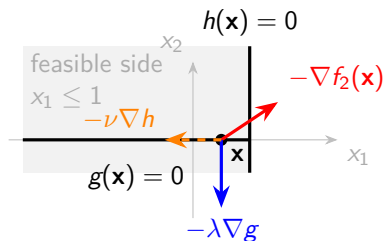
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Quick review—One common “physical” system

$$f_c(\mathbf{x}) = \frac{1}{2}(x_1 - c)^2 - x_2, \quad g(\mathbf{x}) = x_2 = 0, \quad h(\mathbf{x}) = x_1 - 1 \leq 0.$$

- f_c : potential energy of a horizontal spring anchored at $x_1 = c$, together with a constant upward load.
- $g = 0$: a frictionless horizontal rail.
- $h \leq 0$: a rigid wall at $x_1 = 1$; the particle must stay to its left.



Balance law: at $\mathbf{x}^*$, with $\lambda^* \in \mathbf{R}^m$, $\nu^* \in \mathbf{R}_+^p$,

$$\nabla f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(\mathbf{x}^*) + \sum_{j=1}^p \nu_j^* \nabla h_j(\mathbf{x}^*) = \mathbf{0}.$$

Quick review—KKT first-order conditions

- KKT first-order conditions at $\mathbf{x}^*$ with multipliers $(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$:

$$\nabla f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(\mathbf{x}^*) + \sum_{j=1}^p \nu_j^* \nabla h_j(\mathbf{x}^*) = \mathbf{0},$$

$$g_i(\mathbf{x}^*) = 0, \quad i = 1, \dots, m; \quad h_j(\mathbf{x}^*) \leq 0, \quad j = 1, \dots, p,$$

$$\nu_j^* \geq 0, \quad j = 1, \dots, p,$$

$$\nu_j^* h_j(\mathbf{x}^*) = 0, \quad j = 1, \dots, p.$$

Complementary slackness: contact status.

Quick review—KKT first-order conditions

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$$\begin{aligned}\nabla f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(\mathbf{x}^*) + \sum_{j=1}^p \nu_j^* \nabla h_j(\mathbf{x}^*) &= \mathbf{0}, \\ g_i(\mathbf{x}^*) &= 0, \quad i = 1, \dots, m; \quad h_j(\mathbf{x}^*) \leq 0, \quad j = 1, \dots, p, \\ \nu_j^* &\geq 0, \quad j = 1, \dots, p, \\ \nu_j^* h_j(\mathbf{x}^*) &= 0, \quad j = 1, \dots, p.\end{aligned}$$

Complementary slackness: contact status.

- LICQ: a regular local force environment (**not solely for points of interest**).

$$\{\nabla g_i(\mathbf{x}) : i = 1, \dots, m\} \cup \{\nabla h_j(\mathbf{x}) : j \in J(\mathbf{x})\}$$

is linearly independent, where $J(\mathbf{x}) := \{j : h_j(\mathbf{x}) = 0\}$ is the active set.

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- Local minimizer + LICQ $\Rightarrow$ KKT FONC ($(\boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ exist and are unique).
What about the reverse direction?

Trap 1: KKT FONC are not sufficient

$$\min_{x_1, x_2} f(\mathbf{x}) = -x_1^2, \quad \text{s. t.} \quad h(\mathbf{x}) = x_2 = 0.$$

At $\mathbf{x}^* = \mathbf{0}$, LICQ holds, and

$$\nabla f(\mathbf{x}^*) + 0 \cdot \nabla h(\mathbf{x}^*) = \mathbf{0}.$$

But $\mathbf{x}^* = \mathbf{0}$ is a global maximizer.

First-order balance cannot distinguish a stable minimum from an unstable equilibrium.

Trap 2: KKT FONC can hold without LICQ

$$\min_{\mathbf{x} \in \mathbb{R}^2} -x_1 + x_2^2 \quad \text{s. t.} \quad h_1(\mathbf{x}) = x_1 \leq 0, \quad h_2(\mathbf{x}) = 2x_1 \leq 0.$$

The origin is the unique global minimizer. Both constraints are active, but

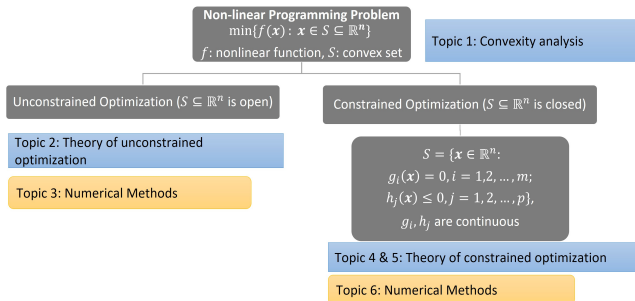
$$\nabla h_2(\mathbf{0}) = 2\nabla h_1(\mathbf{0}),$$

so LICQ fails. Nevertheless the KKT FONC hold whenever

$$-1 + \nu_1 + 2\nu_2 = 0, \quad \nu_1, \nu_2 \geq 0.$$

LICQ guarantees existence and **uniqueness** of multipliers satisfying the KKT FONC at a local minimizer. Its failure does not imply that those conditions must fail.

Topics



In Topic 4, we study optimality conditions for **constrained optimization**.

- KKT first-order necessary conditions and constraint qualifications
- KKT second-order necessary and sufficient conditions
- KKT first-order necessary and sufficient conditions for convex optimization and SVM

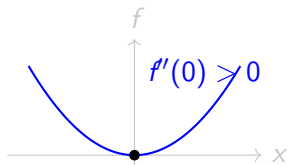
KKT: Karush–Kuhn–Tucker.

Recap: unconstrained second-order tests

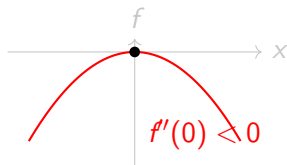
At a stationary point $\nabla f(\mathbf{x}^*) = \mathbf{0}$:

$$\mathbf{x}^* \text{ local minimum} \implies \mathbf{H}_f(\mathbf{x}^*) \succeq 0,$$

$$\mathbf{H}_f(\mathbf{x}^*) \succ 0 \implies \mathbf{x}^* \text{ strict local minimum.}$$



positive curvature: stable



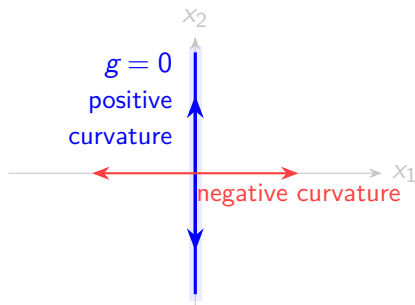
negative curvature: unstable

Constraints remove irrelevant directions

Consider

$$f(x_1, x_2) = -x_1^2 + x_2^2, \quad g(\mathbf{x}) = x_1 = 0.$$

The full Hessian is indefinite, but the feasible line sees only positive curvature.



Which *part* of the feasible directions must pass the curvature test?

First-order-neutral directions: the critical cone

Suppose that $\mathbf{x}^*$ is a local minimizer where LICQ holds. Recall the linearized feasible-direction cone

$$T_{\text{lin}}(\mathbf{x}^*) = \left\{ \mathbf{d} : \begin{array}{l} \nabla g_i(\mathbf{x}^*)^\top \mathbf{d} = 0, \\ \nabla h_j(\mathbf{x}^*)^\top \mathbf{d} \leq 0, \quad j \in J(\mathbf{x}^*) \end{array} \right\} = T_S(\mathbf{x}^*).$$

First-order-neutral directions: the critical cone

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Definition 13.1 (Critical cone)

$$C_{\text{crit}}(\mathbf{x}^*) := \{ \mathbf{d} \in T_{\text{lin}}(\mathbf{x}^*) : \nabla f(\mathbf{x}^*)^\top \mathbf{d} = 0 \}.$$

That is, the linearized feasible directions on which first order is **silent**.

Given a KKT triple: the first equivalent form of C_{crit}

$$C_{\text{crit}}(\mathbf{x}^*) := \{\mathbf{d} \in T_{\text{lin}}(\mathbf{x}^*) : \nabla f(\mathbf{x}^*)^T \mathbf{d} = 0\}.$$

For $\mathbf{d} \in T_{\text{lin}}(\mathbf{x}^*)$, KKT first-order stationarity gives

$$\nabla f(\mathbf{x}^*)^T \mathbf{d} = - \sum_{j \in J(\mathbf{x}^*)} \nu_j^* \nabla h_j(\mathbf{x}^*)^T \mathbf{d} \geq 0.$$

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$$\nabla f(\mathbf{x}^*)^T \mathbf{d} = - \sum_{j \in J(\mathbf{x}^*)} \nu_j^* \nabla h_j(\mathbf{x}^*)^T \mathbf{d} \geq 0.$$

Hence

$$C_{\text{crit}}(\mathbf{x}^*) = \{\mathbf{d} \in T_{\text{lin}}(\mathbf{x}^*) : \nu_j^* \nabla h_j(\mathbf{x}^*)^T \mathbf{d} = 0, \quad j \in J(\mathbf{x}^*)\}.$$

Given a KKT triple: the second equivalent form of C_{crit}

$$C_{\text{crit}}(\mathbf{x}^*) := \{\mathbf{d} \in T_{\text{lin}}(\mathbf{x}^*) : \nabla f(\mathbf{x}^*)^\top \mathbf{d} = 0\}$$

$\Downarrow$ KKT

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Split the active set according to the load:

$$J_+ := \{j \in J(\mathbf{x}^*) : \nu_j^* > 0\}, \quad J_0 := \{j \in J(\mathbf{x}^*) : \nu_j^* = 0\}.$$

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Then

$$C_{\text{crit}}(\mathbf{x}^*) = \left\{ \mathbf{d} : \begin{array}{ll} \nabla g_i(\mathbf{x}^*)^\top \mathbf{d} = 0, & i = 1, \dots, m, \\ \nabla h_j(\mathbf{x}^*)^\top \mathbf{d} = 0, & j \in J_+, \\ \nabla h_j(\mathbf{x}^*)^\top \mathbf{d} \leq 0, & j \in J_0. \end{array} \right\}.$$

Example: compute a critical cone

$$\min_{x_1, x_2} f(\mathbf{x}) = x_1 + x_2^2 \quad \text{s. t.} \quad h_1(\mathbf{x}) = -x_1 \leq 0, \quad h_2(\mathbf{x}) = -x_2 \leq 0.$$

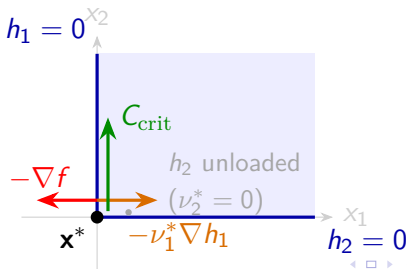
At $\mathbf{x}^* = \mathbf{0}$, $f(\mathbf{x}) \geq 0$ on the feasible first quadrant, so $\mathbf{0}$ is the unique global minimizer. KKT stationarity gives

$$(1, 0)^T + \nu_1^*(-1, 0)^T + \nu_2^*(0, -1)^T = \mathbf{0}, \quad (\nu_1^*, \nu_2^*) = (1, 0).$$

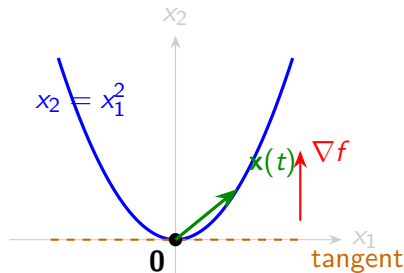
Therefore

$$T_{\text{lin}}(\mathbf{0}) = \{\mathbf{d} : d_1 \geq 0, d_2 \geq 0\},$$

$$C_{\text{crit}}(\mathbf{0}) = \{(0, d_2)^T : d_2 \geq 0\}.$$



Does the (ambient) Hessian of f suffice?



$$\begin{array}{ll} \min & f(\mathbf{x}) = x_2 \\ \text{s. t.} & g(\mathbf{x}) = x_2 - x_1^2 = 0 \end{array}$$

$\mathbf{H}_f = 0$, yet along $\mathbf{x}(t) = (t, t^2)$,

$$f(\mathbf{x}(t)) = t^2.$$

Constraint curvature creates the positive second-order growth.

Hessian of Lagrangian, not only Hessian of f

Fix $\mathbf{d} \in C_{\text{crit}}(\mathbf{x}^*)$ for a KKT triple $(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$, and assume LICQ at $\mathbf{x}^*$. Then a controlled smooth feasible curve $\mathbf{x}(t) \in \mathcal{S}$ can be chosen with

$$\mathbf{x}(0) = \mathbf{x}^*, \quad \mathbf{x}'(0) = \mathbf{d}, \quad \mathbf{x}''(0) = \mathbf{w},$$

and $g_i(\mathbf{x}(t)) = 0$, $h_j(\mathbf{x}(t)) = 0$ for every $j \in J_+$ (loaded).

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Then

$$\left. \frac{d^2}{dt^2} f(\mathbf{x}(t)) \right|_{t=0} = \mathbf{d}^T \mathbf{H}_f(\mathbf{x}^*) \mathbf{d} + \nabla f(\mathbf{x}^*)^T \mathbf{w}.$$

The acceleration $\mathbf{w}$ records how the path must bend to remain feasible. In general, it cannot be discarded if $\nabla f(\mathbf{x}^*) \neq \mathbf{0}$.

Hessian of Lagrangian, not only Hessian of f (cont'd)

$$\mathbf{x}(0) = \mathbf{x}^*, \quad \mathbf{x}'(0) = \mathbf{d}, \quad \mathbf{x}''(0) = \mathbf{w}.$$

$$g_i(\mathbf{x}(t)) = 0, \quad h_j(\mathbf{x}(t)) = 0, \quad \forall j \in J_+.$$

$$\left. \frac{d^2}{dt^2} f(\mathbf{x}(t)) \right|_{t=0} = \mathbf{d}^T \mathbf{H}_f(\mathbf{x}^*) \mathbf{d} + \nabla f(\mathbf{x}^*)^T \mathbf{w}.$$

KKT stationarity condition: $\exists \boldsymbol{\lambda}^* \in \mathbf{R}^m, \boldsymbol{\nu}^* \in \mathbf{R}_+^p$, such that

$$\nabla f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i^* \nabla g_i(\mathbf{x}^*) + \sum_{j \in J_+} \nu_j^* \nabla h_j(\mathbf{x}^*) = \mathbf{0}.$$

What to do with $\nabla g_i(\mathbf{x}^*)^T \mathbf{w}$ and $\nabla h_j(\mathbf{x}^*)^T \mathbf{w}$?

Hessian of Lagrangian, not only Hessian of f (cont'd)

$$\begin{aligned}\left. \frac{d^2}{dt^2} f(\mathbf{x}(t)) \right|_{t=0} &= \mathbf{d}^\top \mathbf{H}_f(\mathbf{x}^*) \mathbf{d} + \nabla f(\mathbf{x}^*)^\top \mathbf{w} \\ &= \mathbf{d}^\top \mathbf{H}_f(\mathbf{x}^*) \mathbf{d} - \sum_{i=1}^m \lambda_i^* \nabla g_i(\mathbf{x}^*)^\top \mathbf{w} - \sum_{j \in J_+} \nu_j^* \nabla h_j(\mathbf{x}^*)^\top \mathbf{w}.\end{aligned}$$

Along the controlled path, twice differentiating

$$g_i(\mathbf{x}(t)) = 0 \quad \text{and} \quad h_j(\mathbf{x}(t)) = 0, \quad j \in J_+$$

gives

$$\begin{aligned}\nabla g_i(\mathbf{x}^*)^\top \mathbf{w} + \mathbf{d}^\top \mathbf{H}_{g_i}(\mathbf{x}^*) \mathbf{d} &= 0, \\ \nabla h_j(\mathbf{x}^*)^\top \mathbf{w} + \mathbf{d}^\top \mathbf{H}_{h_j}(\mathbf{x}^*) \mathbf{d} &= 0, \quad j \in J_+.\end{aligned}$$

Hessian of Lagrangian, not only Hessian of f (cont'd)

$$\begin{aligned}\left. \frac{d^2}{dt^2} f(\mathbf{x}(t)) \right|_{t=0} &= \mathbf{d}^T \mathbf{H}_f(\mathbf{x}^*) \mathbf{d} + \nabla f(\mathbf{x}^*)^T \mathbf{w} \\ &= \mathbf{d}^T \mathbf{H}_f(\mathbf{x}^*) \mathbf{d} + \mathbf{d}^T \left(\sum_{i=1}^m \lambda_i^* \mathbf{H}_{g_i}(\mathbf{x}^*) + \sum_{j \in J_+} \nu_j^* \mathbf{H}_{h_j}(\mathbf{x}^*) \right) \mathbf{d}\end{aligned}$$

Hessian of Lagrangian, not only Hessian of f (cont'd)

$$\begin{aligned}\left. \frac{d^2}{dt^2} f(\mathbf{x}(t)) \right|_{t=0} &= \mathbf{d}^T \mathbf{H}_f(\mathbf{x}^*) \mathbf{d} + \nabla f(\mathbf{x}^*)^T \mathbf{w} \\ &= \mathbf{d}^T \mathbf{H}_f(\mathbf{x}^*) \mathbf{d} + \mathbf{d}^T \left(\sum_{i=1}^m \lambda_i^* \mathbf{H}_{g_i}(\mathbf{x}^*) + \sum_{j \in J_+} \nu_j^* \mathbf{H}_{h_j}(\mathbf{x}^*) \right) \mathbf{d} \\ &= \mathbf{d}^T \mathbf{H}_f(\mathbf{x}^*) \mathbf{d} + \mathbf{d}^T \left(\sum_{i=1}^m \lambda_i^* \mathbf{H}_{g_i}(\mathbf{x}^*) + \sum_{j=1}^p \nu_j^* \mathbf{H}_{h_j}(\mathbf{x}^*) \right) \mathbf{d}\end{aligned}$$

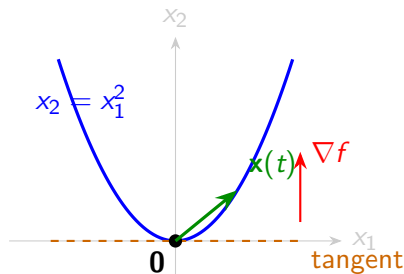
Hessian of Lagrangian, not only Hessian of f (cont'd)

$$\begin{aligned}\left. \frac{d^2}{dt^2} f(\mathbf{x}(t)) \right|_{t=0} &= \mathbf{d}^T \mathbf{H}_f(\mathbf{x}^*) \mathbf{d} + \nabla f(\mathbf{x}^*)^T \mathbf{w} \\ &= \mathbf{d}^T \mathbf{H}_f(\mathbf{x}^*) \mathbf{d} + \mathbf{d}^T \left(\sum_{i=1}^m \lambda_i^* \mathbf{H}_{g_i}(\mathbf{x}^*) + \sum_{j \in J_+} \nu_j^* \mathbf{H}_{h_j}(\mathbf{x}^*) \right) \mathbf{d} \\ &= \mathbf{d}^T \mathbf{H}_f(\mathbf{x}^*) \mathbf{d} + \mathbf{d}^T \left(\sum_{i=1}^m \lambda_i^* \mathbf{H}_{g_i}(\mathbf{x}^*) + \sum_{j=1}^p \nu_j^* \mathbf{H}_{h_j}(\mathbf{x}^*) \right) \mathbf{d} \\ &= \mathbf{d}^T \mathbf{H}_L \mathbf{d}.\end{aligned}$$

where

$$\mathbf{H}_L := \nabla_{\mathbf{xx}}^2 L(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) = \mathbf{H}_f(\mathbf{x}^*) + \sum_{i=1}^m \lambda_i^* \mathbf{H}_{g_i}(\mathbf{x}^*) + \sum_{j=1}^p \nu_j^* \mathbf{H}_{h_j}(\mathbf{x}^*).$$

Returning to the example



$$\begin{array}{ll} \min & f(\mathbf{x}) = x_2 \\ \text{s. t.} & g(\mathbf{x}) = x_2 - x_1^2 = 0 \end{array}$$

$\mathbf{H}_f = 0$, yet along $\mathbf{x}(t) = (t, t^2)$,

$$f(\mathbf{x}(t)) = t^2.$$

$$\nabla f(\mathbf{0}) = (0, 1)^T, \nabla g(\mathbf{0}) = (0, 1)^T, \lambda^* = -1, C_{\text{crit}}(\mathbf{0}) = \{(d_1, 0)^T : d_1 \in \mathbf{R}\}$$

$$\mathbf{H}_L = \mathbf{H}_f(\mathbf{0}) + \lambda^* \mathbf{H}_g(\mathbf{0}) = \begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}.$$

For any $\mathbf{d} = (d_1, 0)^T \in C_{\text{crit}}(\mathbf{0})$, $\mathbf{d}^T \mathbf{H}_L \mathbf{d} = 2d_1^2 \geq 0$.

KKT second-order necessary condition (SONC)

Theorem 13.2

Suppose the data are C^2 , $\mathbf{x}^$ is a local minimizer, and LICQ holds at $\mathbf{x}^*$. For its unique KKT multipliers,*

$$\mathbf{d}^T \mathbf{H}_L \mathbf{d} \geq 0 \quad \forall \mathbf{d} \in C_{\text{crit}}(\mathbf{x}^*).$$

Proof. Fix $\mathbf{d} \in C_{\text{crit}}(\mathbf{x}^*)$ and use the controlled feasible path from LICQ. Set $\phi(t) := f(\mathbf{x}(t))$. Then

$$\phi(t) \geq \phi(0), \quad \phi'(0) = \nabla f(\mathbf{x}^*)^T \mathbf{d} = 0.$$

Therefore

$$0 \leq \phi''(0) = \mathbf{d}^T \mathbf{H}_L \mathbf{d}.$$

□

Special case: unconstrained optimization, $\mathbf{H}_L = \mathbf{H}_f(\mathbf{x}^*)$, $C_{\text{crit}}(\mathbf{x}^*) = \mathbf{R}^n$.

How to use the KKT SONC

- 1 Verify LICQ; find a feasible KKT candidate and its multipliers.
- 2 Identify J_+ and J_0 ; construct $C_{\text{crit}}(\mathbf{x}^*)$.
- 3 Form $\mathbf{H}_L = \nabla_{\mathbf{x}\mathbf{x}}^2 L(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$.
- 4 Search for $\mathbf{d} \in C_{\text{crit}}(\mathbf{x}^*)$ with $\mathbf{d}^T \mathbf{H}_L \mathbf{d} < 0$.

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- 3 Form $\mathbf{H}_L = \nabla_{\mathbf{x}\mathbf{x}}^2 L(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$.
- 4 Search for $\mathbf{d} \in C_{\text{crit}}(\mathbf{x}^*)$ with $\mathbf{d}^T \mathbf{H}_L \mathbf{d} < 0$.

When $C_{\text{crit}}(\mathbf{x}^*)$ is a linear subspace, let the columns of $\mathbf{Z}$ form a basis, so $C_{\text{crit}} = \text{Range}(\mathbf{Z})$. Then

$$\mathbf{d}^T \mathbf{H}_L \mathbf{d} \geq 0 \quad \forall \mathbf{d} \in C_{\text{crit}} \iff \mathbf{Z}^T \mathbf{H}_L \mathbf{Z} \succeq 0.$$

$\mathbf{Z}^T \mathbf{H}_L \mathbf{Z}$ is also called the **reduced Hessian**.

When does the critical cone become a subspace?

$$C_{\text{crit}}(\mathbf{x}^*) = \left\{ \mathbf{d} : \begin{array}{ll} \nabla g_i(\mathbf{x}^*)^\top \mathbf{d} = 0, & i = 1:m, \\ \nabla h_j(\mathbf{x}^*)^\top \mathbf{d} = 0, & j \in J_+, \\ \nabla h_j(\mathbf{x}^*)^\top \mathbf{d} \leq 0, & j \in J_0. \end{array} \right\}.$$

There are two common (nontrivial) cases.

- ① **Equality constraints only:** $C_{\text{crit}} = \{\mathbf{d} : \nabla g_i(\mathbf{x}^*)^\top \mathbf{d} = 0\}.$
- ② **Strict complementarity:** $h_j(\mathbf{x}^*) = 0 \Rightarrow \nu_j^* > 0$, so $J_0 = \emptyset$ and

$$C_{\text{crit}} = \{\mathbf{d} : \nabla g_i^\top \mathbf{d} = 0, \nabla h_j^\top \mathbf{d} = 0 \ (j \in J)\}.$$

Reduced-Hessian example

$$\min_{x_1, x_2} f(\mathbf{x}) = x_1^2 - x_2^2 + x_3^2 \quad \text{s. t.} \quad g(\mathbf{x}) = x_1 = 0.$$

LICQ holds at $\mathbf{x}^* = \mathbf{0}$,

$$\nabla f(\mathbf{0}) = \mathbf{0}, \quad \nabla g(\mathbf{0}) = (1, 0, 0)^T, \quad \lambda^* = 0.$$

$$\mathbf{H}_L = \begin{pmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \quad \mathbf{z} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad \mathbf{z}^T \mathbf{H}_L \mathbf{z} = \begin{pmatrix} -2 & 0 \\ 0 & 2 \end{pmatrix} \not\equiv 0.$$

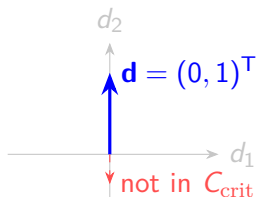
When the critical cone is only a cone

$$\min f(\mathbf{x}) = x_1 - x_2^2 \quad \text{s. t.} \quad -x_1 \leq 0, \quad -x_2 \leq 0.$$

At $\mathbf{x}^* = \mathbf{0}$, LICQ holds, and the KKT first-order conditions give

$$\boldsymbol{\nu}^* = (1, 0)^T, \quad J_+ = \{1\}, \quad J_0 = \{2\}.$$

$C_{\text{crit}}(\mathbf{0}) = \{(0, d_2)^T : d_2 \geq 0\}$ is a ray, not a subspace.



$$\mathbf{H}_L = \begin{pmatrix} 0 & 0 \\ 0 & -2 \end{pmatrix}.$$

Test directly on the cone:

$$\mathbf{d}^T \mathbf{H}_L \mathbf{d} = -2 < 0.$$

KKT second-order sufficient condition (SOSC)

Theorem 13.3

Suppose the data are C^2 and $(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*)$ satisfies the KKT first-order conditions. If

$$\mathbf{d}^\top \mathbf{H}_L \mathbf{d} > 0 \quad \forall \mathbf{d} \in C_{\text{crit}}(\mathbf{x}^*) \setminus \{\mathbf{0}\},$$

then there exist $c, \varepsilon > 0$ such that

$$f(\mathbf{x}) \geq f(\mathbf{x}^*) + c\|\mathbf{x} - \mathbf{x}^*\|^2 \quad (\mathbf{x} \in \mathcal{S}, \|\mathbf{x} - \mathbf{x}^*\| < \varepsilon).$$

Hence $\mathbf{x}^*$ is a strict local minimizer.

NB. Here no LICQ is required.

Special cases: unconstrained optimization ($\mathbf{H}_L = \mathbf{H}_f(\mathbf{x}^*)$, $C_{\text{crit}}(\mathbf{x}^*) = \mathbf{R}^n$).

SOSC success 1: curvature supplied by a constraint

$$\min f(\mathbf{x}) = x_2 \quad \text{s. t.} \quad g(\mathbf{x}) = x_2 - x_1^2 = 0.$$

At $\mathbf{0}$, $\nabla f(\mathbf{0}) = (0, 1)^\top$, $\nabla g(\mathbf{0}) = (0, 1)^\top$, $\lambda^* = -1$ and

$$C_{\text{crit}} = \{(d_1, 0)^\top : d_1 \in \mathbb{R}\}, \quad \mathbf{H}_L = \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix}.$$

For every nonzero critical direction,

$$\mathbf{d}^\top \mathbf{H}_L \mathbf{d} = 2d_1^2 > 0.$$

SOSC certifies a strict local minimum, exactly as the feasible restriction $f(x_1, x_1^2) = x_1^2$ predicts.

SOSC success 2: an indefinite Hessian can work on a cone

$$\min f(\mathbf{x}) = \frac{1}{4}(x_1^2 + x_2^2) + x_1x_2 \quad \text{s. t.} \quad h_1(\mathbf{x}) = -x_1 \leq 0, \quad h_2(\mathbf{x}) = -x_2 \leq 0.$$

At $\mathbf{0}$, $\nabla f(\mathbf{0}) = \mathbf{0}$, $\nabla h_1(\mathbf{0}) = (-1, 0)^\top$, $\nabla h_2(\mathbf{0}) = (0, -1)^\top$, $\nu^* = \mathbf{0}$ and $C_{\text{crit}} = \mathbb{R}_+^2$. Moreover,

$$\mathbf{H}_L = \begin{pmatrix} 1/2 & 1 \\ 1 & 1/2 \end{pmatrix} \quad \text{has eigenvalues } \frac{3}{2} \text{ and } -\frac{1}{2},$$

but for $\mathbf{0} \neq \mathbf{d} \in \mathbb{R}_+^2$,

$$\mathbf{d}^\top \mathbf{H}_L \mathbf{d} = \frac{1}{2}(d_1^2 + d_2^2) + 2d_1d_2 > 0.$$

SOSC succeeds because only the feasible cone matters.

KKT SONC is necessary but not sufficient

$$\min_{x_1, x_2} f(\mathbf{x}) = x_2^3 \quad \text{s. t.} \quad x_1 = 0.$$

At $\mathbf{x}^* = \mathbf{0}$, the KKT first-order conditions hold, and the critical cone is $C_{\text{crit}} = \{(0, d_2)^T : d_2 \in \mathbb{R}\}$, and

$$\mathbf{d}^T \mathbf{H}_L \mathbf{d} = 0 \quad \forall \mathbf{d} \in C_{\text{crit}}.$$

Thus SONC holds, but $f(0, x_2) = x_2^3$ changes sign near zero.

KKT SOSC is sufficient but not necessary

$$\min_{x_1, x_2} x_2^4 \quad \text{s. t.} \quad x_1 = 0.$$

The origin is the strict global minimizer on the feasible line, but

$$C_{\text{crit}} = \{(0, d_2)^T : d_2 \in \mathbb{R}\}, \quad \mathbf{d}^T \mathbf{H}_L \mathbf{d} = 0 \quad \forall \mathbf{d} \in C_{\text{crit}}.$$

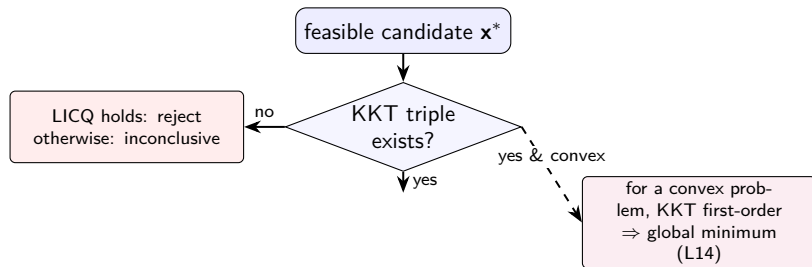
SOSC fails because the first nonzero growth is fourth order.

KKT second-order conditions: filter versus certificate

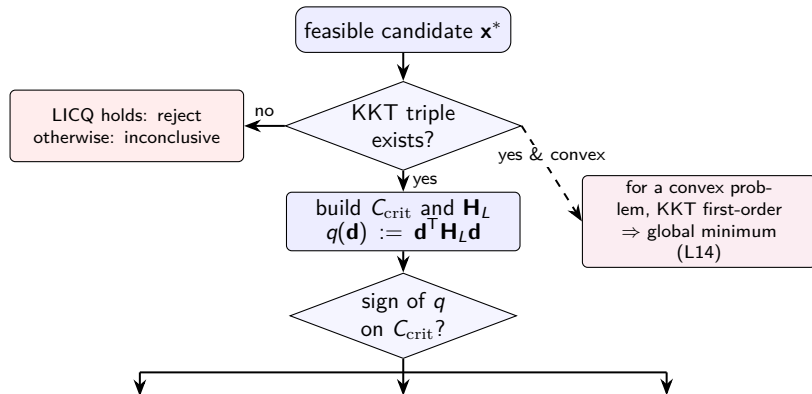
Test	Curvature on C_{crit}	Conclusion
SONC	$\mathbf{d}^T \mathbf{H}_L \mathbf{d} \geq 0$ for all $\mathbf{d}$	necessary at a local minimizer under LICQ
SOSC	$\mathbf{d}^T \mathbf{H}_L \mathbf{d} > 0$ for all $\mathbf{0} \neq \mathbf{d}$	quadratic growth; strict local minimum

- A negative critical direction rejects a regular candidate.
- Positive critical curvature certifies strict local minimality.
- A zero-curvature critical direction leaves a higher-order question.

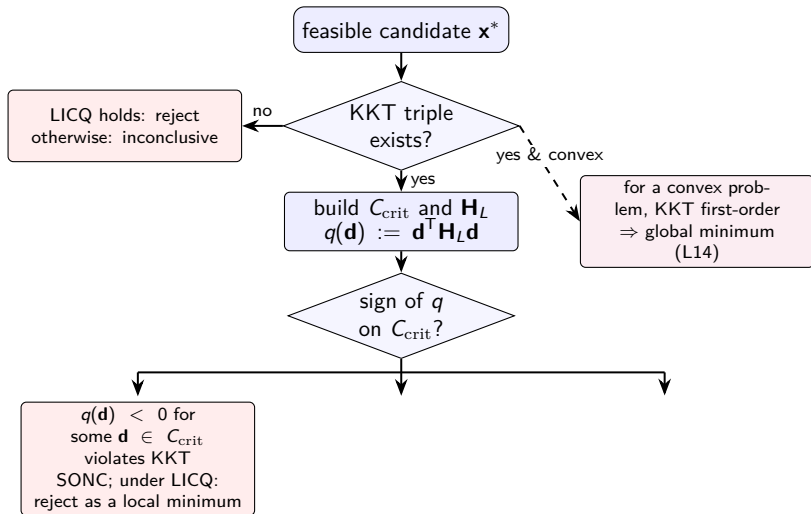
KKT first-order + second-order workflow



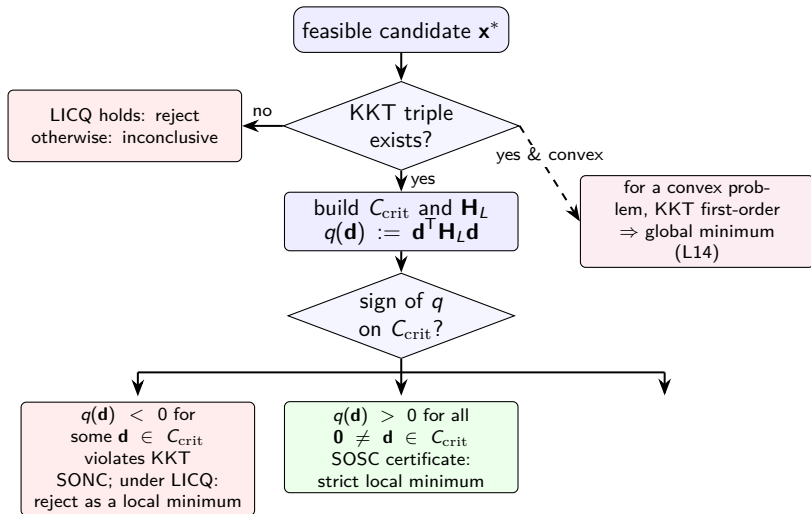
KKT first-order + second-order workflow



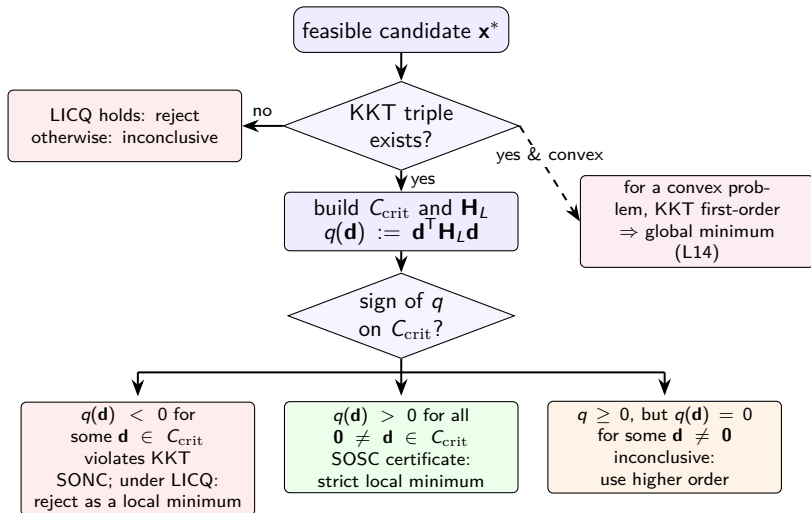
KKT first-order + second-order workflow



KKT first-order + second-order workflow



KKT first-order + second-order workflow



Take-home messages

- C_{crit} : linearized feasible directions that first order cannot classify.
- In C_{crit} : loaded $\Rightarrow$ equalities, unloaded $\Rightarrow$ inequalities.
- $\mathbf{H}_L$, not $\mathbf{H}_f$: feasible-path bending.
- Filter: local minimizer + LICQ $\Rightarrow$ KKT SONC; no “ $\Leftarrow$ ”.
- Certificate: local minimizer $\Leftarrow$ KKT triple + SOSC; no “ $\Rightarrow$ ”.
- C_{crit} is a subspace $\Rightarrow$ reduced Hessians.

Appendix: realizing a critical direction

For $\mathbf{d} \in C_{\text{crit}}$, let $J(\mathbf{d}) = \{j \in J : \nabla h_j(\mathbf{x}^*)^\top \mathbf{d} = 0\}$. Under LICQ, the gradients of g_i and h_j , $j \in J(\mathbf{d})$, are linearly independent. The inverse function theorem produces a C^2 curve satisfying

$$g_i(\mathbf{x}(t)) = 0, \quad h_j(\mathbf{x}(t)) = 0 \ (j \in J(\mathbf{d})), \quad \mathbf{x}'(0) = \mathbf{d}.$$

For $j \in J \setminus J(\mathbf{d})$,

$$h_j(\mathbf{x}(t)) = t \nabla h_j(\mathbf{x}^*)^\top \mathbf{d} + o(t) < 0$$

for small $t > 0$; inactive inequalities remain strict by continuity. Thus the curve is feasible and keeps every constraint in $J_+ \subseteq J(\mathbf{d})$ active.

Appendix: the complete $\mathbf{H}_L$ calculation

With $\mathbf{w} = \mathbf{x}''(0)$,

$$\begin{aligned}(f \circ \mathbf{x})''(0) &= \mathbf{d}^\top \mathbf{H}_f \mathbf{d} + \nabla f^\top \mathbf{w} \\&= \mathbf{d}^\top \mathbf{H}_f \mathbf{d} - \sum_i \lambda_i^* \nabla g_i^\top \mathbf{w} - \sum_{j \in J_+} \nu_j^* \nabla h_j^\top \mathbf{w} \\&= \mathbf{d}^\top \left(\mathbf{H}_f + \sum_i \lambda_i^* \mathbf{H}_{g_i} + \sum_{j \in J_+} \nu_j^* \mathbf{H}_{h_j} \right) \mathbf{d} \\&= \mathbf{d}^\top \mathbf{H}_L \mathbf{d}.\end{aligned}$$

The second equality uses stationarity; the third uses the twice-differentiated constraint equalities along the controlled path.

Appendix: why KKT SOSC needs no constraint qualification

For a feasible point $\mathbf{x}$ and any KKT multipliers,

$$L(\mathbf{x}, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) \leq f(\mathbf{x}), \quad L(\mathbf{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\nu}^*) = f(\mathbf{x}^*).$$

Thus a hypothetical sequence violating quadratic growth can be studied directly by Taylor expanding L , whose $\mathbf{x}$ -gradient vanishes at $\mathbf{x}^*$.

Normalized feasible displacements automatically yield a nonzero $\mathbf{d} \in C_{\text{crit}}$ with $\mathbf{d}^T \mathbf{H}_L \mathbf{d} \leq 0$, contradicting SOSC. No realization of arbitrary linearized directions is needed.

Appendix: reduced-Hessian basis invariance

If $\hat{\mathbf{Z}} = \mathbf{Z}\mathbf{R}$ is another basis of the same critical subspace, with $\mathbf{R}$ nonsingular, then

$$\hat{\mathbf{Z}}^T \mathbf{H}_L \hat{\mathbf{Z}} = \mathbf{R}^T (\mathbf{Z}^T \mathbf{H}_L \mathbf{Z}) \mathbf{R}.$$

Congruence preserves positive semidefiniteness and positive definiteness. Therefore the reduced-Hessian conclusion is independent of the chosen basis.